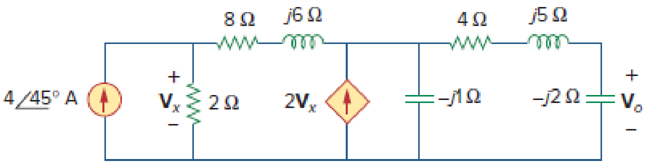
Chapter 10, Problem 18P

Problem

Use nodal analysis to obtain **V***o* in the circuit of Fig. 10.67 below.

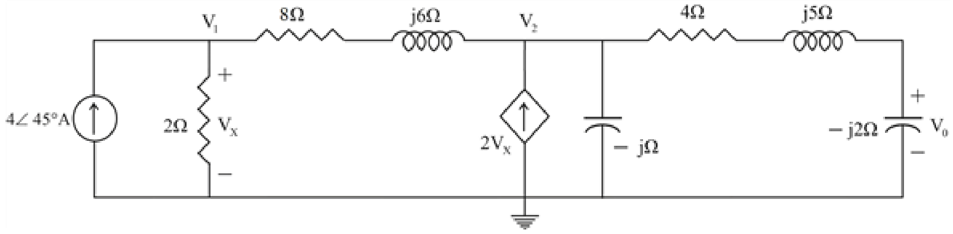
Figure 10.67:



Step-by-step solution

Step 1 of 5

Consider the circuit shown below.



Step 2 of 5

At node 1,

$$4\angle 45° = \frac{V_1}{2} + \frac{V_1 - V_2}{8 + j6}$$

$$4\angle 45° = \frac{1}{2}\left[\frac{V_1}{1} + \frac{V_1 - V_2}{4 + j3}\right]$$

$$8\angle 45° = V_1\left[1 + \frac{1}{4 + j3}\right] - \frac{V_2}{4 + j3}$$

$$8\angle 45° = V_1\left[\frac{5 + j3}{4 + j3}\right] - \left(\frac{V_2}{4 + j3} \times \frac{4 - j3}{4 - j3}\right)$$

$$8\angle 45° = V_1\left[\frac{5 + j3}{4 + j3}\right] - \left(\frac{V_2(4 - j3)}{25}\right)$$

$$8\angle 45° = V_1[1.16 - j0.12] - \left(\frac{V_2(4 - j3)}{25}\right)$$

$$200\angle 45° = (29 - j3)V_1 - (4 - j3)V_2 \quad \text{..............(1)}$$

Step 3 of 5

At node 2,

$$\frac{V_1 - V_2}{8 + j6} + 2V_X = \frac{V_2}{-j} + \frac{V_2}{4 + j5 - j2}$$

Where, $V_X = V_1$

$$\frac{V_1 - V_2}{8 + j6} + 2V_1 = jV_2 + \frac{V_2}{4 + j3}$$

$$V_1\left[\frac{1}{8 + j6} + 2\right] = V_2\left[j + \frac{1}{4 + j3} + \frac{1}{8 + j6}\right]$$

$$V_1\left[\frac{1 + 16 + j12}{8 + j6}\right] = V_2\left[j + \frac{1}{2}\left(\frac{2}{4 + j3} + \frac{1}{4 + j3}\right)\right]$$

$$V_1\left[\frac{17 + j12}{8 + j6}\right] = V_2\left[j + \frac{1}{2}\left(\frac{3}{4 + j3}\right)\right]$$

$$V_1\left[\frac{17 + j12}{8 + j6}\right] = V_2\left[\left(\frac{8j - 6 + 3}{8 + j6}\right)\right]$$

$$V_1\left[\frac{17 + j12}{8 + j6}\right] = V_2\left[\left(\frac{-3 + j8}{8 + j6}\right)\right]$$

$$V_1[17 + j12] = V_2[-3 + j8]$$

$$V_1 = V_2\left[\frac{-3 + j8}{17 + j12}\right] \text{................ (2)}$$

Step 4 of 5

Substituting (2) into (1)

$$200\angle 45^\circ = (29 - j3)\left[\frac{-3 + j8}{17 + j12}\right]V_2 - (4 - j3)V_2$$

$$200\angle 45^\circ = [4.205 + j11.207]V_2 - (4 - j3)V_2$$

$$200\angle 45^\circ = [(4.205 + j11.207) - (4 - j3)]V_2$$

$$200\angle 45^\circ = [0.205 + j14.207]V_2$$

$$200\angle 45^\circ = (14.21\angle 89.17^\circ)V_2$$

$$\therefore V_2 = \frac{200\angle 45^\circ}{14.21\angle 89.17^\circ}$$

Step 5 of 5

Therefore,

$$V_0 = \left(\frac{-j2}{4+j5-j2}\right)V_2$$

$$V_0 = \left(\frac{-j2}{4+j3}\right)V_2$$

$$V_0 = \left(0.4\angle -126.87^\circ\right)V_2$$

$$V_0 = \left(0.4\angle -126.87^\circ\right)\times\left(\frac{200\angle 45^\circ}{14.21\angle 89.17^\circ}\right)$$

$$V_0 = \left(0.4\angle -126.87^\circ\right)\times\left(14.07\angle -44.17^\circ\right)$$

$$\therefore \boxed{V_0 = 5.628\angle -171.04^\circ\ \text{V}}$$

Or,

$$\therefore \boxed{V_0 = 5.628\angle 189^\circ\ \text{V}}$$